

P100-023: Finger-pad friction during viscous liquid spreading

Principia-100 | Standardized portfolio | 1 October 2026

Scenario and evaluation contract

Domain: tribology and biomechanics. The current default task predicts Author-derived median coefficient of friction in 1. Independent unit: whole participant. Its packaged cohort contains 30 assigned rows in 2 groups.

All supplied outcomes are now exposed. Historical reserved evaluations remain part of the evidence record; future scores are retrospective. No newly established fundamental law or measured deployment impact is admitted by this packaging.

Current portfolio findings

Reproduction: An established Hersey/Stribeck response is the useful original reference; physiological additions do not establish a new rule.

Limit: Source full-data exponent 0.55 and fitted rheology are shared prior information, not independent verification of the same mechanism.

Retrospective extension: A train-only free-exponent pressure model improves same-fluid person transfer but fails the stronger unseen-fluid transfer challenge.

Limit: Negative pressure exponent appears 9/9 folds but Hersey itself contains inverse force. Photographic area is not instantaneous loaded area; fixed fluid order and shared calibration undermine causal pressure interpretation.

Versioned tasks

Task family	Target and information	Source scope
original	Author-derived median coefficient of friction (1)	original corpus
continuation	Author-derived median coefficient of friction (1)	original corpus

Different targets, calibration budgets and cohorts have different task IDs. Their raw error values cannot be pooled into a scenario ranking. Exact equations, coefficients, permitted variables, timing, groups and per-model results are bound in the task packages.

Evaluation: the shared CLI provides numerical scoring, explicit coverage and abstention, uncertainty and event diagnostics. Agent review separately assesses scientific support; no aggregate discovery score is produced.

Detailed reference note: Finger-liquid friction: a person-transfer rule with fluid limits

Research edition: 1 October 2026. Historical results remain preserved; this reader edition uses the shared benchmark evaluator.

Supported findings and the competing explanation

The original source reports a full Stribeck collapse and its high-H exponent 0.55. This continuation fits both branch exponents only on the original development participants, removing that source fitted exponent advantage. The pressure-conditioned candidate is

$$\hat{\mu} = a + bh^{-p}\pi^r + ch^q, \quad h = H/0.02, \quad \pi = (F_N/1N)/(A/4\text{cm}^2).$$

Here A is photographic pad area, not the instantaneous loaded contact area. Frozen coefficients are $a=2.5790\text{e-}11$, $b=0.126335$, $c=0.0405567$, $p=0.263105$, $q=0.754523$, $r=-0.322758$. At fixed author H, the boundary contribution decreases with normalized force/area. The sign is negative in 9/9 participant-fold fits, ranging -0.454 to -0.163. However H itself contains inverse force; these observational data do not identify causal pressure physics.

Whole-person development MAE is 0.077064 versus 0.078925 without pressure and 0.104467 with fixed original exponents. The exposed two-person diagnostic gives 0.084917, versus 0.097362 fixed-exponent and 0.081720 kernel control. The kernel has lower diagnostic error; it is not promoted after exposure.

The stricter development-only challenge removes both a person and the tested fluid from training. Pressure MAE rises to 0.120469, versus 0.090762 without pressure. This falsifies a transferable fluid-independent pressure law. The free-exponent curve is the better structural explanation for unseen-fluid development stress, but the original preselected same-fluid candidate is preserved. Fixed 500/s viscosity Hersey also loses to the shear-thinning source Hersey, reproducing the importance of rheology calibration rather than discovering it.

Value, limits and next evidence

The value is a more honest source-exponent comparison and a concrete example of participant-transfer gains failing fluid transfer. Hydration additions do not improve the main development task. Fixed fluid order, one substrate, trial-median target and shared fluid calibrations prevent prospective formulation or causal skin-state claims. No new universal friction law or deployed benefit is admitted. Randomized repeated fluid trials and independent contact geometry are needed before promotion.

Reproduction and evaluation

Run `python run.py` to verify frozen predictions. Use `python evaluator/evaluate.py example --output /tmp/submission` and `python evaluator/evaluate.py score --submission /tmp/submission --output /tmp/report --trust-code` in a new output location. This trusted-code replay is not a security sandbox. Evaluator scores exposed targets and does not grant novelty from error. Exact source/sample anchors are in `data/observations.csv.gz`; complete numerical evidence and unfavorable models remain in `evidence/`.

Sources and prior art

- [Native record](#)
- [Original Stribeck study](#)

Targeted primary-source checking is not exhaustive novelty adjudication. Established model-family success is distinguished from a new physical law.